6.3.6 Oropharyngeal Cancer

◆ Key Features

- Oropharyngeal cancer includes cancer of the palatine tonsil, tongue base, soft palate, and oropharyngeal wall.
- Oropharyngeal cancer is primarily linked to tobacco and alcohol use.
- Human papillomavirus (HPV) is a risk factor for cancer of the tonsil.
- Neck metastasis may be cystic.

The oropharynx is located between the soft palate superiorly and the hyoid bone inferiorly; it communicates with the oral cavity anteriorly, the nasopharynx superiorly, and the supraglottic larynx and hypopharynx inferiorly. Oropharyngeal cancers are typically detected at a more advanced stage than oral cancer. The oropharynx is an important component in swallowing; therefore, treating these tumors is challenging and often requires a multidisciplinary approach and posttreatment rehabilitation.

◆ Epidemiology

In the United States, an estimated 8,300 new cases of pharyngeal cancer (including cancers of the oropharynx and hypopharynx) are diagnosed yearly, with an estimated mortality of 2,000. It affects men three times more than women. Seventy-five percent of oropharynx cancers occur in the palatine tonsil. Tobacco (including smokeless tobacco) and alcohol abuse represent the most significant risk factors for the development of oropharynx cancer. Viral infection with HPV is an important risk factor for squamous cell carcinoma (SCC) of the oropharynx and may be a positive prognostic factor.

◆ Clinical

Signs

Signs include changes in articulation, muffled speech, a mass in the neck, unintentional weight loss, hemoptysis, and persistent halitosis.

Symptoms

Symptoms may include pain, dysphagia, globus sensation, referred otalgia, trismus, and fixation of the tongue. Cancers of the base of the tongue and tonsil are typically insidious.

Differential Diagnosis

- Oropharyngeal infections such as pharyngitis or stomatitis
- Chancre
- Benign oropharyngeal or odontogenic lesions
- Aphthous ulcers or herpetic sores
- Oral manifestations of systemic diseases

◆ Evaluation

History

Evaluation begins with a detailed history inquiring about tobacco and alcohol usage, sexual history, oral pain, odynophagia, referred otalgia, dysphagia, hemoptysis, articulation or speech changes, and unintentional weight loss.

Physical Exam

The physical exam should include a complete head and neck exam, with specific attention directed at the site of the lesion. The lesion size should be noted, as should its apparent infiltration and spread to adjacent pharyngeal or oral cavity subsites such as oral tongue, hypopharynx, nasopharynx, and vallecula. Palpation of the lesion should be performed if possible in an awake patient. In advanced cases, the primary origin of the lesion, such as the tongue base or tonsil, is not always discernible. Fiberoptic laryngoscopy should be performed. Inspection and palpation of the neck often reveal adenopathy.

Imaging

Clinical staging may understage oropharynx tumors, especially the tongue base extension. Both contrast-enhanced computed tomography (CT) and magnetic resonance imaging (MRI) are able to assess oropharynx tumor extent as well as regional spread. Neck metastasis from oropharynx cancer may be cystic in morphology; this finding by itself should raise suspicion of a cancer in the tonsil or tongue base.

At minimum, a chest CT should be used to rule out pulmonary metastasis or second primary. Alternatively, an integrated ^{18}F fludeoxyglucose (FDG)–positron emission tomography (PET)/CT may be used in a combined staging and metastatic work-up.

Labs

Standard preoperative laboratories, as indicated, should be obtained.

Other Tests

Patients with suspected cancer of oropharynx must undergo a biopsy and a sample of the lesion taken for pathologic examination. This may be done in an office setting in cases of tonsil cancer and soft palate cancer, but it is not usually possible in cases of tongue base lesions. If neck metastases are evident, they should be sampled by fine-needle aspiration biopsy (FNAB).

Because of the propensity for second primary tumors, panendoscopy (triple endoscopy—laryngoscopy, esophagoscopy, and bronchoscopy together) has been advocated. This is particularly important in patients who smoke or in patients with large, bulky tumors to establish the true extent of these lesions.

Pathology

Histologically, 90% of oropharynx cancers are SCCs. Basaloid SCC is an uncommon but aggressive SCC variant. Other cancers of the oropharynx include minor salivary gland carcinomas, lymphomas, and “lymphoepithelial-like” carcinomas.

◆ Treatment Options

For stage I oropharynx cancer, surgery or radiotherapy may be used depending on the expected functional deficit. Radiation clinical trials evaluating hyperfractionation schedules should be considered.

For stage II oropharynx cancer, surgery and radiation therapies are equally successful in controlling disease. Radiotherapy may be the preferred modality when the functional deficit is expected to be great.

The management of stage III oropharynx cancer is complex and requires a multidisciplinary approach to establish the optimal treatment. A combination of surgery with postoperative radiotherapy and/or chemotherapy is most often used. An alternative is chemoradiation therapy alone, based on the patient's initial nodal involvement.

In stage III tonsil cancer, hyperfractionated radiotherapy yields a higher control rate than standard fractionated radiotherapy.

In advanced unresectable oropharyngeal cancer, radiotherapy or chemoradiation is used.

Treatments currently under investigation include chemotherapy with radiation clinical trials as well as with radiosensitizers, radiation clinical trials evaluating hyperfractionation schedules and/or brachytherapy, particle-beam radiotherapy, and hyperthermia combined with radiotherapy.

Recently, transoral robotic-assisted surgery (TORS) for resection of select oropharyngeal tumors has gained momentum. Clinical trials are currently underway to evaluate the outcomes of TORS versus radiation therapy.

◆ Outcome and Follow-Up

The overall 5-year disease-specific survival for patients with all stages of disease is ~ 50%. Patients with HPV-positive tumors who are nonsmokers have a better prognosis. The National Comprehensive Cancer Network (NCCN) guidelines suggest routine follow up after treatment. This includes follow up every 1 to 2 months the first year, every 2 to 6 months the second year, every 4 to 8 months the third through fifth years and annual follow up afterwards.

◆ Staging of Oropharyngeal Cancer (HPV Negative)

Primary Tumor (T)

- TX: Primary tumor cannot be assessed
- Tis: Carcinoma in situ
- T1: Tumor ≤ 2 cm in greatest dimension
- T2: Tumor > 2 cm but ≤ 4 cm in greatest dimension
- T3: Tumor > 4 cm in greatest dimension or extension to lingual surface of epiglottis
- T4: Moderately advanced or very advanced local disease
  - T4a: Moderately advanced local disease. Tumor invades the larynx, extrinsic muscle of tongue, medial pterygoid, hard palate, or mandible.*
  - T4b: Very advanced local disease. Tumor invades lateral pterygoid muscle, pterygoid plates, lateral nasopharynx or skull base or encases carotid artery.

*Note: Mucosal extension to the lingual surface of the epiglottis from primary tumors of the base of the tongue and vallecula does not constitute invasion of the larynx.

Regional Lymph Nodes (N)*

- NX: Regional lymph nodes cannot be assessed
- N0: No regional lymph node metastasis
- N1: Metastasis to a single ipsilateral lymph node measuring ≤ 3 cm in greatest diameter and ENE negative
- N2: Further divided into three categories:
  - N2a: Single ipsilateral lymph node > 3 and ≤ 6 cm and ENE negative (pathologic staging also includes a single ipsilateral or contralateral lymph node ≤ 3 cm and ENE positive)
  - N2b: Multiple ipsilateral lymph nodes ≤ 6 cm and ENE negative
  - N2c: Bilateral or contralateral lymph nodes ≤ 6 cm in greatest dimension and ENE negative
- N3: Now further divided into two categories:
  - N3a: Metastasis to a lymph node > 6 cm, ENE negative
  - N3b: Metastasis to a single lymph node that is ENE positive (pathologic staging includes single lymph nodes that are ENE positive >3cm), or metastasis to multiple lymph nodes with any ENE positive

A designation of "U" or "L" can be attached to the N category to indicate nodes above ("U") or below ("L") the lower border of the cricoid. Similarly, clinical and pathologic ENE should be recorded as ENE (–) or ENE (+). Clinical and pathologic staging are similar with the differences noted in parenthesis.

*Superior mediastinal lymph nodes are considered regional lymph nodes (level VII). Midline nodes are considered ipsilateral nodes.

Please refer to the next chapter for staging of HPV positive oropharyngeal cancer.

6.3.7 Human Papillomavirus and Head and Neck Cancer

◆ Key Features

- HPV-related oropharyngeal cancer is increasing in incidence.
- Between 70 and 90% of newly diagnosed oropharyngeal cancers in the United States are HPV-positive.
- Patients with HPV-related head and neck cancers tend to be younger and healthier and have an improved prognosis compared to those with HPV-negative tumors.
- Neck metastasis may be cystic.

Human papillomavirus (HPV) is the most common sexually transmitted infection. Infections can occur in the oral cavity (as well as in the cervix or anorectal or urogenital tract). Persistent high-risk HPV infection (primarily HPV16) is associated with development of malignancy. Most HPV infections clear within one year spontaneously and do not require treatment. Some infections may become persistent, especially in immunocompromised patients or active smokers. Routine HPV testing of oropharyngeal squamous cell carcinoma and head and neck squamous cell cancers of unknown primary are recommended. Testing should include high-risk HPV in situ hybridization (ISH) and/or p16 immunohistochemistry (IHC).

Discussing HPV may be unfamiliar territory in otolaryngology—head and neck surgery practice, and it can be a potentially sensitive topic that can be challenging to navigate with patients. Patients and partners have questions and concerns regarding acquisition and transmission of HPV infection, but partners of patients with HPV-related oropharyngeal cancer do not have a higher rate of oral HPV infection.

◆ Epidemiology

HPV is the most common sexually transmitted infection in the United States, and it is estimated that a majority of the sexually active population will have at least one HPV infection during their lifetimes. HPV is a double-stranded DNA virus that can infect human skin and mucosa, including the oropharynx, genitals, and anus. More than 150 types of HPV infections have been identified, and these can be divided into low-risk and high-risk types. Low-risk infections, most commonly HPV6 and 11, are associated with development of benign lesions such as genital warts or respiratory papillomatosis. High-risk infections, most commonly HPV16 and 18, are associated with development of malignancies including cervical, anorectal, urogenital, and oropharyngeal cancer.

In a patient with a cancer of unknown primary, the tumors frequently are found to be HPV-positive.

Pathophysiology

The mechanism of viral oncogenesis is through HPV viral proteins E6 and E7, which disrupt cell cycle pathways. The E6 protein binds and degrades p53, allowing cell cycle escape, and E7 inhibits another tumor suppressor, retinoblastoma (Rb) protein.

◆ Clinical

HPV-positive tumors often present with small primary oropharyngeal tumors (**Fig. 6.13**). Nodal disease may be large and cystic in nature.

◆ Treatment Options

Current guidelines do not differentiate treatment recommendations based on HPV tumor status. Patients are treated with chemotherapy, radiation, transoral robotic surgery (TORS), open surgery, or a combination of these. HPV-positive tumor status may serve as eligibility for certain clinical trials investigating de-escalation of treatment.

◆ Outcomes and Follow-Up

Despite late stage at diagnosis, HPV-positive patients have significantly improved prognosis compared to HPV-negative patients. Three-year overall survival is 82% for HPV-positive patients.

◆ Staging of HPV+ Oropharyngeal Cancer

Primary Tumor (T)

T0: No primary identified
T1: Tumor ≤ 2 cm in greatest dimension
T2: Tumor > 2 cm but ≤ 4 cm in greatest dimension
T3: Tumor > 4 cm in greatest dimension or extension to lingual surface of epiglottis
T4: Moderately advanced local disease; tumor invades the larynx, extrinsic muscle of tongue, medial pterygoid, hard palate, or mandible or beyond*

*Mucosal extension to lingual surface of epiglottis from primary tumors of the base of the tongue and vallecula does not constitute invasion of the larynx.

Regional Lymph Node (N)

Clinical N (cN)

NX: Regional lymph nodes cannot be assessed
N0: No regional lymph node metastasis
N1: One or more ipsilateral lymph nodes, ≤ 6 cm
N2: Contralateral or bilateral lymph nodes, ≤ 6 cm
N3: Lymph node(s) > 6 cm

Pathologic N (pN)

NX: Regional lymph nodes cannot be assessed
pN0: No regional lymph node metastasis
pN1: Metastasis in ≤ 4 lymph nodes
pN2: Metastasis in > 4 lymph nodes

Distant Metastasis (M)

M0: No distant metastasis
M1: Distant metastasis

American Joint Committee on Cancer Stage Groupings for HPV positive Oropharyngeal Cancers (8th Edition)

Clinical staging is as follows:

Stage I disease: T0-T2, N0-N1 disease that is M0
Stage II disease: T3N0-N1 M0 and T0-T3 disease that is N2 M0
Stage III disease: T4 tumors with or without nodal disease as well as any tumor with N3 disease without metastases

Stage IV disease: any tumor with metastatic disease

Please refer to **Table 6.12**.

Table 6.12 Clinical staging of HPV + oropharynx cancer

	N0	N1	N2	N3
T0	N/A	Stage I	Stage II	Stage III
T1	Stage I	Stage I	Stage II	Stage III
T2	Stage I	Stage I	Stage II	Stage III
T3	Stage II	Stage II	Stage II	Stage III
T4	Stage III	Stage III	Stage III	Stage III
M1	Stage IV	Stage IV	Stage IV	Stage IV

Staging based on pathologic criteria for HPV associated (p16-positive) oropharyngeal cancer is different and is as follows:

Stage I disease: T0-T2, N0-N1 disease that is M0

Stage II disease: T3-T4N0-N1 M0 and T0-T2 disease that is N2 M0

Stage III disease: T3-T4 with N2M0 disease

Stage IV disease: any tumor with metastatic disease

Please refer to **Table 6.13**.

Table 6.13 Pathologic staging of HPV + oropharynx cancer

	N0	N1	N2
T0	N/A	Stage I	Stage II
T1	Stage I	Stage I	Stage II
T2	Stage I	Stage I	Stage II
T3	Stage II	Stage II	Stage III
T4	Stage II	Stage II	Stage III
M1	Stage IV	Stage IV	Stage IV

Histologic Grade (G)

No grading system exists for HPV-mediated oropharyngeal tumors.

Histopathologic Type

The histopathology of HPV-mediated p16+ oropharyngeal cancers is characteristic and easily recognizable (**Fig. 6.13**).

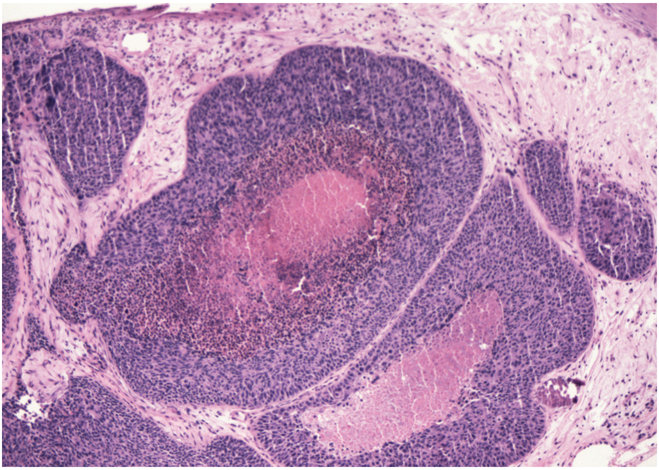

Fig. 6.13 Basaloid squamous cell carcinomas are aggressive tumors, often arranged as nests of small dark cells with central necrosis. Tumor cells at the periphery of the nests frequently align with elongated nuclei at right angles to the stroma, giving a palisaded appearance. When located in the oropharynx, these tumors often harbor human papillomavirus. (Used with permission from Har-El G, Day T, Nathan C-A, Nguyen SA. *A Multidisciplinary Approach to Head and Neck Neoplasms.* New York: Thieme;2013. *Otorhinolaryngology—Head and Neck Surgery Series.*)

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6.3.10 Laryngeal Cancer

◆ Key Features

- Laryngeal cancer is the second most common head and neck cancer.
- The most common site is the glottis.
- It may present early with hoarseness.
- Treatment options are based on site and extent of disease.

Cancer arising from the squamous epithelium of the larynx is a common head and neck cancer, with well-known risk factors. Proper staging is complex but crucial for treatment formulation. The larynx plays a unique role in respiration, speech, and swallowing. The complex anatomy of the larynx explains the unique patterns of spread of laryngeal cancer:

- The preepiglottic fat is located in the anterior and lateral aspects of the larynx and is often invaded by advanced cancers.
- The recurrent laryngeal nerve innervates the intrinsic laryngeal muscles. Invasion of this nerve causes hoarseness clinically and fixation of the vocal folds.
- The anterior commissure invasion may be a conduit for tumor spread.

The larynx is divided into three anatomic regions: the supraglottic larynx, the glottis, and the subglottic region. The supraglottic larynx includes the epiglottis, the preepiglottic space, the laryngeal aspects of the aryepiglottic folds, the false vocal folds, the arytenoids, and the ventricles. The inferior boundary is a horizontal plane drawn through the apex of the ventricle. This corresponds to the area of transition from squamous to respiratory epithelium. The glottis extends from the true vocal folds inferiorly to roughly 1 cm below the true folds, including the paraglottic space and the anterior and posterior commissures extending inferiorly ~ 1 cm. The subglottic larynx has its superior border at the inferior border of the glottis, ~ 1 cm below the true vocal folds, and extends inferiorly to the trachea.

Laryngeal cancer can also be classified by anatomic location. Signs, symptoms, and tumor behavior vary depending on the site and the extent of disease.

◆ Epidemiology

For 2016, the Surveillance, Epidemiology, and End Results (SEER) program of the National Cancer Institute, US National Institutes of Health, estimated that 13,340 men and women would be diagnosed with cancer of the larynx in the United States, with 3,620 patient deaths. Risk factors include smoking and drinking alcohol, which act synergistically; laryngeal papillomatosis; radiation exposure; immunosuppression; and occupational exposure to metals, plastics, and asbestos. Laryngeal carcinoma is more common in African Americans than in European Americans, with a ratio of 3.5:1.

◆ Clinical

Signs and Symptoms

The common symptoms of *supraglottic cancers* include mild odynophagia, mild dysphagia, and mass sensation. Later symptoms include severe dysphagia and aspiration and referred ear pain (otalgia). The mechanism of the referred ear pain is through the activation of the internal branch of the superior laryngeal branch of cranial nerve (CN) X with referral to Arnold's nerve (auricular branch of CN X). Twenty-five to 50% have nodal metastasis to the cervical lymph nodes. Supraglottic tumors often spread to nodes on both sides of the neck.

Glottic cancers account for over half of all laryngeal cancers and present typically with hoarseness. Voice change often happens early and can help diagnose early-stage cancer, improving prognosis. Additionally, the vocal folds' sparse lymphatics leads to a relatively low incidence of nodal metastasis. Advancing glottic cancer can spread posteriorly to the arytenoid complex, causing vocal fold fixation, or anteriorly to the commissure, where it can invade the thyroid cartilage.

Subglottic cancer is uncommon. It may produce biphasic stridor or airway obstruction. Patients may be asymptomatic until advanced stages of disease, and thus the prognosis is worse. Lymphatics drain through the cricothyroid and cricotracheal membranes to the pretracheal, paratracheal, and inferior jugular nodes, and occasionally to mediastinal nodes.

Differential Diagnosis

- Hyperkeratosis
- Papillomas
- Polyps
- Fibromas
- Granulomas
- Laryngoceles
- Laryngeal manifestations of systemic, infectious, or autoimmune disease

◆ Evaluation

The most important adverse prognostic factors for laryngeal cancers include increasing T stage and N stage (see the "Staging of Laryngeal Cancer" section). Other prognostic factors may include sex, age, performance status, and a variety of pathologic features of the tumor, including grade and depth of invasion.

History

History should focus on timing and duration of symptoms and assessment of risk factors. A full understanding of the patient's general medical health, nutritional and cardiopulmonary status, social support, and compliance are crucial for planning of therapy.

Physical Examination

A thorough physical examination of the head and neck should be performed, including inspection of the oral mucosa, laryngoscopy, bimanual palpation of the floor of the mouth and the base of the tongue, and a careful assessment of the cervical lymph nodes and thyroid cartilage contour.

Imaging

Contrast-enhanced computed tomography (CT) scans obtained with appropriate section thickness aid in the evaluation of laryngeal cancer and neck masses. Positron emission tomography (PET)-CT can assist in identifying metastasis and set a baseline of future follow-up for recurrence. At the very least, a chest CT should be performed to rule out a secondary pulmonary malignancy or metastatic disease.

Other Tests

Suspension direct laryngoscopy provides an opportunity for examination under general anesthesia, palpation, and biopsy. Extent of the tumor and the overall condition of the airway mucosa can be evaluated. Panendoscopy (i.e., laryngoscopy, esophagoscopy, and bronchoscopy together) can be performed to rule out synchronous malignancies.

Fine-needle aspiration biopsy (FNAB) of a neck mass may yield a positive result when the certainty of an enlarged lymph node is in question.

Pathology

Ninety to 95% of laryngeal cancers are squamous cell carcinomas (SCCs). Other less common types of laryngeal malignancies include adenoid cystic carcinoma, with a characteristic indolent course of growth and perineural invasion. Rarely, cancer can be of neuroendocrine, stromal, or mesenchymal origin.

◆ Treatment Options

Stage I Laryngeal Cancer

Supraglottis

Standard treatment options include:

- External-beam radiotherapy alone
- Supraglottic laryngectomy

Glottis

Standard treatment options include:

- Radiotherapy
- Cordectomy for very carefully selected patients with limited and superficial T1 lesions
- Partial or hemilaryngectomy or total laryngectomy, depending on anatomic considerations
- Laser excision (endoscopic laryngectomy)

Subglottis

- Radiotherapy alone

Stage II Laryngeal Cancer

Supraglottis

Standard treatment options include:

- External-beam radiotherapy, supraglottic laryngectomy, or total laryngectomy, depending on location of the lesion, clinical status of the patient, and expertise of the treatment team
- Postoperative radiotherapy, indicated for positive or close surgical margins

Glottis

Standard treatment options include:

- Radiotherapy
- Partial or hemilaryngectomy or total laryngectomy, depending on anatomic considerations; under certain circumstances, laser microsurgery may be appropriate.

Subglottis

- Lesions can be treated successfully by radiotherapy alone with preservation of normal voice.

Stage III Laryngeal Cancer

Supraglottis

Standard treatment options include:

- Surgery with or without postoperative radiotherapy
- Definitive radiotherapy with surgery for salvage of radiation failures
- Chemotherapy administered concomitantly with radiotherapy; can be considered for patients who would require total laryngectomy for control of disease

Glottis

Standard treatment options include:

- Surgery with or without postoperative radiotherapy
- Definitive radiotherapy with surgery for the salvage of radiation failures

- Chemotherapy administered concomitantly with radiotherapy; can be considered for patients who would require total laryngectomy for control of disease

Subglottis

Standard treatment options include:

- Laryngectomy plus isolated thyroidectomy and tracheoesophageal node dissection, usually followed by postoperative radiotherapy
- Treatment by radiotherapy alone, indicated for patients who are not candidates for surgery

Stage IV Laryngeal Cancer

Supraglottis

Standard treatment options include:

- Total laryngectomy with postoperative radiotherapy
- Definitive radiotherapy with surgery for salvage of radiation failures
- Chemotherapy administered concomitantly with radiotherapy; can be considered for patients who would require total laryngectomy for control of disease

Glottis

Standard treatment options include:

- Total laryngectomy with postoperative radiotherapy
- Definitive radiotherapy with surgery for salvage of radiation failures
- Chemotherapy administered concomitantly with radiotherapy; can be considered for patients who would require total laryngectomy for control of disease

Subglottis

Standard treatment options include:

- Laryngectomy plus total thyroidectomy and bilateral tracheoesophageal node dissection, usually followed by postoperative radiotherapy
- Treatment by radiotherapy alone, indicated for patients who are not candidates for surgery

Treatment Types

Radiation

Radiation is sometimes preferred because of the good oncologic results, preservation of the voice, and the possibility of surgical salvage in patients whose disease recurs locally.

Surgery

Surgical treatment options should be reviewed carefully to ensure adequate pulmonary and swallowing function postoperatively.

- *Transoral laser microsurgery:* Transoral laser microsurgery is ideal for the treatment of early to intermediate glottic and supraglottic cancer. It may be performed under suspension microlaryngoscopy with a laser. The tumor may be transected and removed piecemeal. This treatment has the same indications and contraindications as open partial laryngectomies. A functional cricoarytenoid unit must be preserved. Survival and laryngeal preservation are comparable to other conventional treatments.
- *Transoral robotic surgery (TORS):* Transoral robotic-assisted supraglottic laryngectomy may be appropriate for certain supraglottic cancers.
- *Partial laryngectomy:* Several procedures constitute partial laryngectomies:
  - *Vertical hemilaryngectomy:* When disease is small and involves only one vocal fold and arytenoid cartilage, a vertical hemilaryngectomy can be performed. This procedure removes, unilaterally, the true vocal fold from anterior to commissure to vocal process of arytenoid, as well as the false vocal fold, the ventricle paraglottic space, and the overlying thyroid cartilage. Resection can be extended to include the whole arytenoid. A frontolateral hemilaryngectomy additionally removes the anterior commissures, part of the contralateral vocal fold, and the overlying thyroid cartilage.
  - *Supraglottic laryngectomy:* Supraglottic laryngectomy can be performed in tumors limited to the supraglottic region with normal vocal fold mobility, with 2-mm margin at the anterior commissure. It is contraindicated if disease extends anteriorly into the thyroid cartilage or the anterior neck or posteriorly to postcricoid or intra-arytenoid areas or prevertebral fascia, and in patients with poor pulmonary function. This procedure may remove the entire epiglottis and aryepiglottic folds, false vocal folds, preepiglottic space, and part of the thyroid cartilage, and it can be extended to include the vallecula, wall of the piriform recess, and the base of the tongue up to the circumvallate papillae. Supraglottic laryngectomy may be performed open or transorally utilizing a robotic platform.
  - *Supracricoid laryngectomy:* Supracricoid laryngectomy can be performed for removal of large tumors, even certain T4 tumors of the glottis and supraglottis. However, it is contraindicated in subglottic extension, 10 mm anteriorly or 5 mm posteriorly, and with involvement of both arytenoid cartilages, the hyoid bone, the base of the tongue, or the cricoid cartilage. Supracricoid laryngectomy resects both true and both false vocal folds, the paraglottic space, and the entire thyroid cartilage, as well as the epiglottis and one arytenoid if necessary. Reconstruction involves cricohyoidoepiglottopexy or cricohyoidopexy if the epiglottis is removed.
- *Total laryngectomy:* A total laryngectomy may be performed in the event of radiation failure with unclear extent of recurrence, in the event of chondroradionecrosis, or when other conservation techniques are not feasible. A total laryngectomy removes the entire larynx, including the thyroid and cricoid cartilages, part of the thyroid gland, and the paratracheal lymph nodes, and it may involve removal of the overlying strap muscles.

◆ Complications

- Psychosocial trauma from surgery and/or radiotherapy
- Vocal fold–powered voice loss in some procedures
- Aspiration pneumonia, in some procedures
- Osteoradionecrosis
- Chondroradionecrosis
- Chronic pain
- Breathing difficulties
- Stoma infections
- Potential stoma malignancies

◆ Outcome and Follow-Up

Consideration should be given to airway protection, nutritional support, and voice rehabilitation. Establishing a stable airway via tracheotomy should be considered preoperatively for tumors threatening the airway or obscuring intubation. Preemptive gastrostomy tube placement can ensure maximized preoperative and postoperative nutritional status. If voice preservation is not possible in the treatment plan, a noninvasive option for voice rehabilitation is an electrolarynx. Alternately, patients can learn techniques for producing esophageal voice, using expulsion of air through the esophagus to generate the base vibration for speech. With tracheoesophageal puncture, a fistula can be created between the cervical trachea and the esophagus to allow efficient swallowing of air for esophageal voicing.

Close oncologic follow-up care is necessary because second primary cancers, recurrences, and late metastases are all strong possibilities, especially if the patient continues to smoke.

◆ Staging of Laryngeal Cancer

Primary Tumor: Supraglottis (T)

TX: Primary tumor cannot be assessed

Tis: Carcinoma in situ

T1: Tumor limited to one subsite of supraglottis *with normal vocal fold mobility*

T2: Tumor invading mucosa of more than one adjacent subsite of supraglottis or glottis or region outside the supraglottis (e.g., the mucosa of the base of the tongue, vallecula, or medial wall of piriform recess) *without fixation of the larynx*

T3: Tumor limited to larynx *with vocal fold fixation* and/or invading any of the following: the postcricoid area, preepiglottic tissues, paraglottic space, and/or minor thyroid cartilage erosion (e.g., inner cortex)

T4: Moderately advanced to advanced disease

- T4a: Moderately advanced local disease; tumor invading through the thyroid cartilage and/or invading tissues beyond the larynx (e.g., the trachea, the soft tissues of the neck including the deep extrinsic muscles of the tongue, the strap muscles, the thyroid, or the esophagus)

T4b: Very advanced local disease; tumor invading the prevertebral space, encasing the carotid artery, or invading the mediastinal structures

Primary Tumor: Glottis (T)

- TX: Primary tumor cannot be assessed
- Tis: Carcinoma in situ
- T1: Tumor limited to the vocal fold(s) (may involve anterior or posterior commissure) *with normal mobility*
  - T1a: Tumor limited to one vocal fold
  - T1b: Tumor involves both vocal folds
- T2: Tumor extending to the supraglottis and/or the subglottis *and/or with impaired vocal fold mobility*
- T3: Tumor limited to the larynx *with vocal fold fixation* and/or invasion of the paraglottic space, and/or the inner cortex of thyroid cartilage
- T4: Moderately advanced to advanced disease
  - T4a: Tumor invading through the outer cortex of the thyroid cartilage and/or invading tissues beyond the larynx (e.g., the trachea, the soft tissues of the neck including the deep extrinsic muscles of the tongue, strap muscles, the thyroid, or the esophagus)
  - T4b: Tumor invading the prevertebral space, encasing the carotid artery, or invading mediastinal structures

Primary Tumor: Subglottis (T)

- TX: Primary tumor cannot be assessed
- Tis: Carcinoma in situ
- T1: Tumor limited to the subglottis
- T2: Tumor extending to the vocal fold(s) *with normal or impaired mobility*
- T3: Tumor limited to the larynx *with vocal fold fixation*
- T4: Moderately advanced to advanced disease
  - T4a: Tumor invading the cricoid or thyroid cartilage and/or invades the tissues beyond the larynx (e.g., the trachea, the soft tissues of the neck including the deep extrinsic muscles of the tongue, the strap muscles, the thyroid, or the esophagus)
  - T4b: Tumor invading the prevertebral space, encasing the carotid artery, or invading the mediastinal structures

T Category Considerations

- Supraglottis: Normal vocal fold mobility (T1), fixation of the larynx (T2), and vocal fold fixation (T3) may be determined only clinically.
- Glottis: Normal vocal fold mobility (T1), impaired vocal fold mobility (T2), and vocal fold fixation (T3) may be determined only clinically.

- Subglottis: Normal or impaired vocal fold mobility (T2) and vocal fold fixation (T3) may be determined only clinically.

Regional Lymph Nodes (N)*

NX: Regional lymph nodes cannot be assessed

N0: No regional lymph node metastasis

N1: Metastasis to a single ipsilateral lymph node measuring ≤ 3 cm in greatest diameter and ENE negative

N2: Further divided into three categories:

- N2a: Single ipsilateral lymph node > 3 and ≤ 6 cm and ENE negative (pathologic staging also includes a single ipsilateral or contralateral lymph node ≤ 3 cm and ENE positive)
- N2b: Multiple ipsilateral lymph nodes ≤ 6 cm and ENE negative
- N2c: Bilateral or contralateral lymph nodes ≤ 6 cm in greatest dimension and ENE negative

N3: Now further divided into two categories:

- N3a: Metastasis to a lymph node > 6 cm, ENE negative
- N3b: Metastasis to a single lymph node that is ENE positive (pathologic staging includes single lymph nodes that are ENE positive >3cm), or metastasis to multiple lymph nodes with any ENE positive

A designation of "U" or "L" can be attached to the N category to indicate nodes above ("U") or below ("L") the lower border of the cricoid. Similarly, clinical and pathologic ENE should be recorded as ENE (−) or ENE (+). Clinical and pathologic staging are similar with the differences noted in parenthesis.

*Superior mediastinal lymph nodes are considered regional lymph nodes (level VII). Midline nodes are considered ipsilateral nodes.

Distant Metastasis (M)

MX: Distant metastasis cannot be assessed

M0: No distant metastasis

M1: Distant metastasis

American Joint Committee on Cancer Stage Groupings of Supraglottis, Glottis, and Subglottis (8th edition)

Stage 0 disease: Carcinoma in situ (TisN0M0)

Stage I disease: Includes only T1 N0 M0 tumors

Stage II disease: Includes only T2 N0 M0 tumors

Stage III disease: Includes T3 N0 M0 and T1 -3 disease that is N1 M0

Stage IVA disease: Includes T4a disease with N0-N2 M0 disease and T1-T3 disease that is N2 M0

Stage IVB disease: Includes T4b disease with any nodal disease or T1-T4a disease with N3 disease without distant metastases (M0)

Stage IVC disease: Any disease with distant metastases (M1)

See **Table 6.7** in Chapter 6.3.3.

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6.3.12 Referred Otalgia in Head and Neck Disease

◆ Key Features

- Referred otalgia is the sensation of ear pain originating from a source outside the ear.
- If no otic source for otalgia can be identified, then malignancy must be investigated and excluded.

By definition, referred otalgia is the sensation of ear pain originating from a source outside the ear. Many remote anatomic sites share innervations with the ear, and noxious stimuli to these areas may be perceived as otalgia

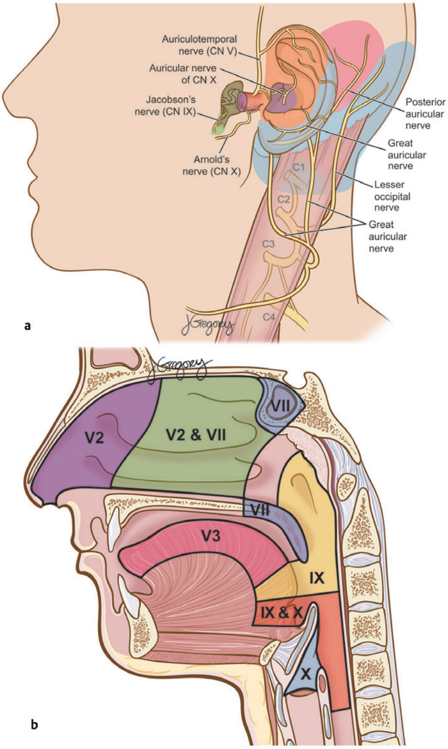

Fig. 6.14 Sources of referred otalgia. **(a)** The sensory innervation of the ear and periauricular area is from multiple nerves, including branches from cranial nerves (CN) V, VII, IX, and X as well as spinal nerve C2 and C3 rootlets. (**b**) Multiple areas in the head and neck share sensory innervation with the ear. (Used with permission from Sclafani AP, ed. *Total Otolaryngology—Head and Neck Surgery*. New York, NY: Thieme;2015:117.)

(**Fig. 6.14**). In adult patients, especially those at risk, if no otic source for otalgia can be identified, malignancy must be investigated and excluded, especially for adult patients with known risk factors.

The sensory innervation of the ear is provided by the auriculotemporal branch of the fifth cranial nerve (CN) (mandibular nerve [CN V_3]), contributions from C2 and C3 through the cervical plexus via the great auricular nerve, the tympanic nerve [branch of CN IX], the auricular nerve [branch of CN X], and the sensory branch of CN VII [Ramsay-Hunt branch], which innervates a portion of the external auditory canal and surrounding concha.

The explanation for complex sensory innervation of the ear ultimately lies in the ear's embryologic development. The otic vesicle comes to rest central to branchial arches 1, 2, 3, and 4. The sensory and motor nerves of these arches are CN V, CN VII, CN IX, and CN X, respectively.

◆ Etiology

Etiology for referred otalgia includes:

- Dental disorders
- Ill-fitting dentures
- Bruxism
- Cervical osteoarthritis
- Cervicofacial myofascial pain syndrome
- Posttraumatic neuralgia (greater auricular neuroma)
- Chronic pharyngitis
- Eagle's syndrome
- Postviral neuralgia (Ramsay-Hunt's syndrome, postherpetic neuralgia)
- Temporal arteritis
- Parotid neoplasia
- Gastroesophageal reflux

Lesions of the anterior two-thirds of the tongue and inferior oral cavity tend to refer pain to the ear via CN V_3. Lesions of the lateral base of the tongue, the tonsillar region, and the inferior two-thirds of the nasopharynx cause deep, intense otalgia via sensory impulses through CN IX. Lesions of the supraglottic larynx communicate sensory afferents via the internal laryngeal branch of the superior laryngeal nerve of CN X. Lesions of the posterior oropharynx, the inferior nasopharynx, the medial aspect of the base of the tongue, and the hypopharynx tend to refer pain to the ear owing to the overlapping innervations of both CN IX and X (via the pharyngeal plexus).

◆ Evaluation

The evaluation of a patient with otalgia begins with a detailed history and a thorough head and neck examination. Questions addressed include the character and timing of the otalgia, exacerbating and alleviating factors of the otalgia, the patient's past otologic history, the symptoms associated with the otalgia (tinnitus, hearing loss, vertigo), the presence of constitutional symptoms (to detect malignancies), and sinus and dental questions.

A head and neck examination is fundamental in evaluating otalgia. A thorough otologic examination, with a tuning fork test at two frequencies (256 and 512 Hz), is important. The CNs are examined and compared bilaterally. The nose, sinuses, oral cavity, oral pharynx, and neck are inspected and palpated to look for sources of referred otalgia. In assessing otalgia in the setting of a normal otologic examination, a fiberoptic nasopharyngolaryngoscopy is mandated to look for lesions that can be potentially noxious to the trigeminal (CN V), facial (CN VII), glossopharyngeal (CN IX), or vagus (CN X) nerves. Attention should be directed to the endolarynx to examine the mucosa for signs of malignancy and gastroesophageal reflux.

◆ Treatment Options

Treatment consists of appropriate treatment of the source of otalgia in combination with pain management.

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6.3.19 Paragangliomas of the Head and Neck

◆ Key Features

- Paragangliomas of the head and neck are highly vascular tumors.
- Other terms used for tumors of the same histology are carotid body tumor, chemodectoma, and glomus tumor.

Paragangliomas are neuroendocrine tumors that develop from paraganglia, small organs of the autonomic nervous system. Paragangliomas may develop at four sites within the head and neck: the carotid body (carotid body [glomus caroticum] tumors), the ganglia of the vagus nerve (glomus vagale tumors), the jugular bulb (glomus jugulare tumors), and the middle ear (glomus tympanicum tumors). Treatment of head and neck paragangliomas is primarily surgical, although radiation and observation may be appropriate under certain circumstances.

◆ Epidemiology

Paragangliomas are rare tumors, accounting for < 1% of all head and neck tumors. In most cases, these are solitary lesions; bilateral or multicentric lesions occur in 3% of cases. Ten to 15% of patients have familial paragangliomas; patients with the familial tumors have much higher rates of bilaterality and multicentricity. The carotid body and jugular bulb are the most common sites of head and neck paragangliomas. Rare sites of involvement are the larynx, nasal cavity, paranasal sinuses, and thyroid gland.

◆ Clinical

Signs and Symptoms

Presentation of paragangliomas relates to the location at which the tumor arises. Carotid body and vagal paragangliomas present as slow-growing, nontender neck masses. A typical lateral neck mass may be seen, accompanied in some cases by the oropharyngeal bulge suggestive of parapharyngeal space involvement. More advanced or neglected lesions may have associated ipsilateral cranial neuropathies.

Differential Diagnosis

Initially, all etiologies of a lateral neck mass should be considered. Radiographic or clinical localization to the parapharyngeal space focuses the investigation; the most common parapharyngeal space lesions are salivary gland tumors (most common), paragangliomas, and nerve sheath tumors (schwannomas and neurofibromas).

◆ Evaluation

Physical Exam

A complete head and neck exam with attention to cranial nerve (CN) function is appropriate. The neck mass associated with a paraganglioma is frequently pulsatile, owing to its intimate relationship to the carotid vessels. Additional neck masses, suggesting lymph node metastasis or multifocality, should also be sought.

Imaging

Computed tomography (CT) scanning with contrast is useful, demonstrating the hypervascular tumor and its relationship with the neck vessels. Magnetic resonance imaging (MRI) provides similar information with the added benefits of no radiation exposure, greater soft tissue differentiation, and three planes of view. Formal angiography has been replaced by MRI, magnetic resonance angiography (MRA), or CT in the diagnostic realm, but it is still required when embolization is needed preoperatively. Octreotide scan is also a helpful nuclear medicine scan that can help confirm a paraganglioma and also identify other potential paragangliomas in the body.

Labs

A history of signs and symptoms of excess catecholamines should be explicitly sought. If symptoms such as flushing, labile or difficult-to-control hypertension, or excessive perspiration are described, preoperative testing for serum and urine catecholamines (and their metabolites) should be ordered. Although “functional” (catecholamine-secreting) paragangliomas of the head and neck are unusual (1–3%), pheochromocytomas of the adrenal medulla, which may constitute a portion of a syndrome of which the head and neck paraganglioma is the presenting feature, are much more frequently metabolically active.

Pathology

Grossly, paragangliomas are polypoid, well circumscribed, and firm to rubbery in consistency. Paragangliomas have a characteristic histologic appearance, with tumor cells arranged in clusters called “Zellballen,” which are separated by a fibrovascular stroma (**Fig. 6.17**). Roughly 10% of paragangliomas are malignant. Malignancy is determined solely on the basis of metastasis rather than on any histologic features of the primary tumor.

◆ Treatment Options

The major goal of treatment is prevention of morbidity due to progressive cranial neuropathies. In most cases, this goal is best achieved by surgical extirpation via a cervical incision. Many surgeons employ preoperative embolization to decrease blood loss and improve visualization with larger

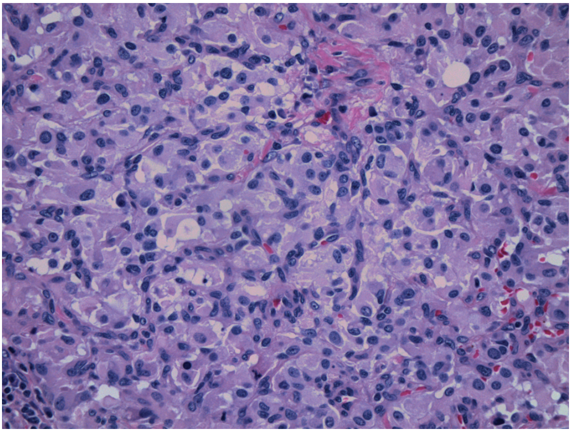

Fig. 6.17 Paraganglioma: a “Zellballen” or compact nested pattern is evident, with plump eosinophilic tumor cells surrounded by spindled S100-positive sustentacular cells. (Used with permission from Har-El G, Day T, Nathan C-A, Nguyen SA. *A Multidisciplinary Approach to Head and Neck Neoplasms*. New York: Thieme;2013. *Otorhinolaryngology—Head and Neck Surgery Series*.)

tumors. The need for internal carotid artery sacrifice and revascularization with saphenous vein grafting should always be considered, especially in larger tumors and tumors that radiographically encapsulate the internal carotid artery. The Shamblin criteria can be used to evaluate the patient, and type III tumors have a higher association with both cranial neuropathies and the need for carotid reconstruction.

On the other hand, surgery itself may result in new cranial neuropathies and significant associated morbidity. This merits consideration of other management strategies in selected cases. Other options include close observation with serial imaging and external beam radiation. Notably, external beam radiation appears to be tumoristatic rather than tumoricidal with paragangliomas.

In the very elderly or infirm patient, and in those with multiple (especially bilateral) tumors, all options are considered carefully. In the fragile patient, surgery may be postponed unless significant growth is noted radiographically or worsening of cranial neuropathies is identified clinically. In patients with bilateral disease, treatment, which may include any combination of surgery, observation, and external beam radiation, is tailored to achieve a goal of CN preservation on at least one side, as intractable swallowing dysfunction and/or tracheotomy dependence are the inevitable consequence of bilateral CN X or XII impairment.

◆ Outcome and Follow-Up

Patients should be closely observed for any local recurrence, although these are usually rare. Contralateral tumors should be sought and resected.